

FINAL MARK

**GIRRAWEE HIGH SCHOOL
MATHEMATICS EXTENSION 1
Task 2 Half Yearly 2005
ANSWERS COVER SHEET**

Name: _____

QUESTION	MARK	HE1	HE2	HE3a	HE3b	HE3c	HE3d	HE4	HE5	HE6	HE7
Question 1	/19	✓									✓
Question 2	/39	✓									✓
Question 3											
a)	/5	✓	✓								✓
b) , c)	/6	✓									✓
d)	/3	✓		✓							✓
e)	/5	✓									✓
Total	/19										
Question 4	/19	✓									✓
TOTAL											
	/96	/96	/5	/3							/96



GIRRAWEE HIGH SCHOOL

**YEAR 12 HALF YEARLY EXAMINATION
Task 2**

2005

**MATHEMATICS
Extension 1**

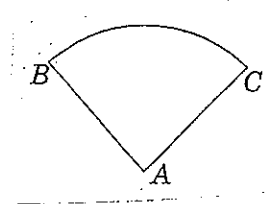
Time allowed - Two hours

(Plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new page.
- A table of standard integrals is provided.

Question 1. (19 marks)

- a) For what values of x is $|x - 1| < x$? 3
- b) Find the term independent of x in the expansion $\left[\frac{1}{x^3} + 2x^5\right]^{16}$ 3
- c) Sketch the following: (i) $y = \tan 2x$ ($0 \leq x \leq 2\pi$) 3
(ii) $y = 3 \sin 2x$ ($-\pi \leq x \leq \pi$) 3
- d) The headlights of a car cast a beam of light in the shape of a sector of a circle, as shown in the diagram. The arc BC of the circle subtends an angle θ at the centre A , with a radius r . The perimeter of the beam ($AB + BC + CA$) is 12 metres.
- 
- Show the area of the sector ABC is given by $A = \frac{720\theta}{(\theta + 2)^2}$ 3
- e) Find the acute angle between $x + 2y - 5 = 0$ and $x - 3y + 3 = 0$. 4

Question 2. (39 marks)

- a) Differentiate:
- (i) $y = \sin^2 3x$ (ii) $y = \tan(\ln 5x)$ 6
- (iii) $y = \frac{e^x}{\tan x}$ (iv) $y = \ln \frac{\sqrt{x}}{x^2 + 1}$ 6
- b) Find the equation of the tangent to the curve $y = -\cos x$ at $x = \frac{\pi}{3}$. 4
- c) Find:
- (i) $\int (\sin x - 4x^3) dx$ (ii) $\int_0^{\frac{\pi}{8}} \sec^2 2t dt$ (iii) $\int_2^3 \frac{3x}{x^2 - 1} dx$ 9
- (iv) $\int_0^1 x^3 e^{2x^4} dx$ (v) $\int_0^{\frac{\pi}{3}} \cos^2 2x dx$ 6
- d) The area enclosed between $y = \frac{2}{\sqrt{x-7}}$ and the x axis between $x = 8$ and $x = 10$ is rotated about the x axis. Find the volume formed. 4
- e) (i) Differentiate $y = \tan(x^2 - 3)$ 2
(ii) Use your result in part (i) to find $\int x \sec^2(x^2 - 3) dx$ 2

Question 3. (19 marks)

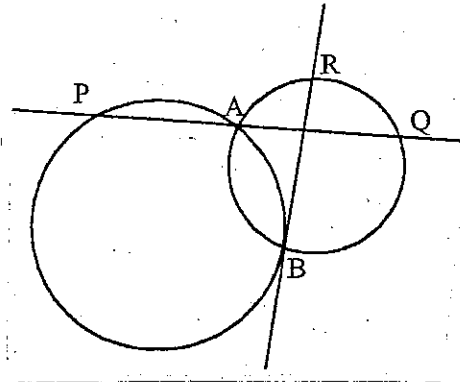
- a) Use Mathematical Induction to prove, $1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$
(for positive integers n) 5
- b) Prove $\frac{\sin \theta}{\sin \theta + \cos \theta} - \frac{\sin \theta}{\sin \theta - \cos \theta} = \tan 2\theta$ 3
- c) Solve $\sin^2 \theta = \frac{1}{2} \sin 2\theta$ ($0 \leq \theta \leq 2\pi$) 3
- d) The probability of a certain family having brown eyed children is $\frac{2}{3}$.
If the family plans to have 4 children, what is the probability that, at most,
two will have brown eyes? Give your answer correct to 2 decimal places. 3
- e) Sketch the curve $y = \frac{x^2 + 1}{x}$, showing all main features, including any asymptotes. 5

Question 4. (20 marks)

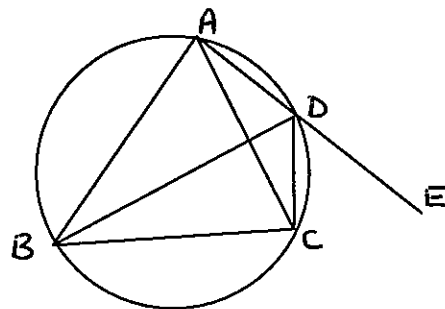
- a) Two circles cut at A and B. A line through A meets one circle at P and the other at Q.

BR is a tangent to circle ABP and R lies on the circle ABQ.

- (i) Copy the diagram
(ii) Prove that $PB \parallel QR$.



- b) ABC is a triangle where $BC = AC$:
(i) Copy the diagram
(ii) Give reasons why $\angle CDE = \angle ABC$ 1
(iii) Hence show that DC bisects $\angle BDE$ 4



- c) (i) On the same set of axes sketch the curves
 $y = 2x + 1$ and $y = \ln(x + 1)$ 3
- (ii) Show that the minimum vertical distance between the curves
 $y = 2x + 1$ and $y = \ln(x + 1)$ occurs when $x = -\frac{1}{2}$ 5

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1.

a) $|x-1| < x$

$\pm(x-1) < x$

$\therefore x-1 < x$ or $-(x-1) < x$

no solution

$x-1 > -x$

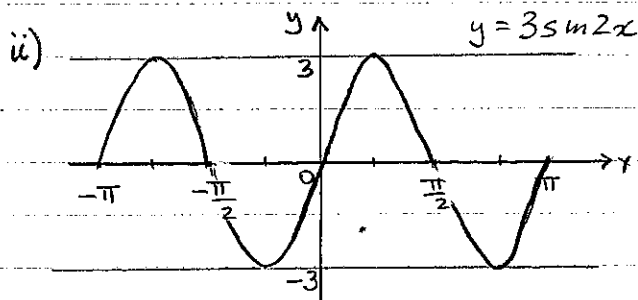
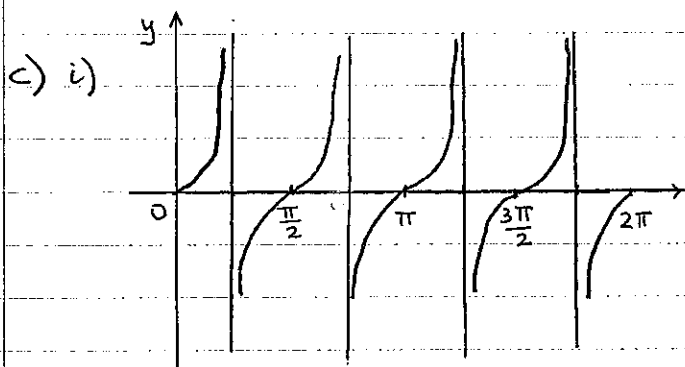
$2x > 1$

$x > \frac{1}{2}$

b) $\left(\frac{1}{x^3} + 2x^5\right)^{16}$

Term is ${}^{16}C_6 \left(\frac{1}{x^3}\right)^{10} (2x^5)^6$

$\therefore$ term is ${}^{16}C_6 \cdot 2^6 (512512)$



d) Arc BC = $r\theta$

Perimeter = $2r + r\theta$

$\therefore 12 = r(2 + \theta)$

$\frac{12}{2 + \theta} = r$

Area = $\frac{1}{2} r^2 \theta$

$= \frac{1}{2} \cdot \frac{144}{(2 + \theta)^2} \cdot \theta$

$A = \frac{72\theta}{(2 + \theta)^2}$

e) $x + 2y - 5 = 0$

$2y = -x + 5$

$y = -\frac{1}{2}x + \frac{5}{2}$ $m_1 = -\frac{1}{2}$

$x - 3y + 3 = 0$

$-3y = -x - 3$

$y = \frac{1}{3}x + 1$ $m_2 = \frac{1}{3}$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{-\frac{1}{2} - \frac{1}{3}}{1 + (-\frac{1}{2}) \cdot \frac{1}{3}} \right|$

$= \left| \frac{-\frac{5}{6}}{1 - \frac{1}{6}} \right|$

$= | -1 |$

$\therefore \tan \theta = 1$

acute angle is 45°

Question 2

a) i) $y = \sin^2 3x$

$\frac{dy}{dx} = 2 \sin 3x \cdot \cos 3x \cdot 3$

$= 6 \sin 3x \cos 3x$

ii) $y = \tan \ln 5x$

$\frac{dy}{dx} = \sec^2 \ln 5x \cdot \frac{1}{5x} \cdot 5$

$= \frac{\sec^2 \ln 5x}{5x}$

$$\text{iii)} \quad y = \frac{e^x}{\tan x}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{\tan x \cdot e^x - e^x \cdot \sec^2 x}{\tan^2 x} \\ &= \frac{e^x (\tan x - \sec^2 x)}{\tan^2 x} \end{aligned}$$

$$\text{iv)} \quad y = \ln\left(\frac{\sqrt{x}}{x^2+1}\right)$$

$$\begin{aligned} y &= \ln \sqrt{x} - \ln(x^2+1) \\ &= \frac{1}{2} \ln x - \ln(x^2+1) \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2x} - \frac{1}{x^2+1} \cdot 2x \\ &= \frac{1}{2x} - \frac{2x}{x^2+1} \end{aligned}$$

$$\text{OR} \quad \frac{1-3x^2}{2x(x^2+1)}$$

$$\text{b)} \quad y = -\cos x \quad x = \frac{\pi}{3}$$

$$y = -\frac{1}{2}$$

$$\frac{dy}{dx} = \sin x$$

$$\text{at } x = \frac{\pi}{3} \quad \frac{dy}{dx} = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

Tangent:

$$y - (-\frac{1}{2}) = \frac{\sqrt{3}}{2} (x - \frac{\pi}{3})$$

$$y + \frac{1}{2} = \frac{\sqrt{3}x}{2} - \frac{\sqrt{3}\pi}{6}$$

$$6y + 3 = 3\sqrt{3}x - \sqrt{3}\pi$$

$$0 = 3\sqrt{3}x - 6y - 3 - \sqrt{3}\pi$$

$$\text{c) i)} \quad \int (\sin x - 4x^3) dx$$

$$= -\cos x - x^4 + C$$

$$\text{ii)} \quad \int_0^{\frac{\pi}{8}} \sec^2 2t dt$$

$$= \left[\frac{\tan 2t}{2} \right]_0^{\pi/8}$$

$$= \frac{\tan \frac{\pi}{4}}{2} - 0$$

$$= \frac{1}{2}$$

$$\text{iii)} \quad \frac{3}{2} \int_2^3 \frac{2x}{x^2-1} dx$$

$$= \left[\frac{3}{2} \ln(x^2-1) \right]_2^3$$

$$= \frac{3}{2} [\ln 8 - \ln 3]$$

$$= \frac{3}{2} \ln \frac{8}{3}$$

$$\text{iv)} \quad \int_0^1 x^3 e^{2x^4} dx = \frac{1}{8} \int_0^1 e^{2x^4} \cdot 8x^3 dx$$

$$= \frac{1}{8} [e^{2x^4}]_0^1$$

$$= \frac{1}{8} [e^2 - e^0]$$

$$= \frac{1}{8} [e^2 - 1]$$

$$\text{v)} \quad \int_0^{\frac{\pi}{3}} \cos^2 2x dx$$

$$\text{Now } \cos 4x = 2\cos^2 2x - 1$$

$$\therefore \cos^2 2x = \frac{1}{2} (\cos 4x + 1)$$

$$\therefore = \frac{1}{2} \int_0^{\pi/3} (\cos 4x + 1) dx$$

$$= \frac{1}{2} \left[\frac{\sin 4x}{4} + x \right]_0^{\pi/3}$$

$$= \frac{1}{2} \left[\frac{\sin \frac{4\pi}{3}}{4} + \frac{\pi}{3} - 0 \right]$$

$$= \frac{1}{2} \left(-\frac{\sqrt{3}}{4} + \frac{\pi}{3} \right)$$

$$= -\frac{\sqrt{3}}{16} + \frac{\pi}{6}$$

$$\begin{aligned}
 \text{d) } V &= \pi \int_8^{10} y^2 dx \\
 &= \pi \int_8^{10} \left(\frac{4}{x-7} \right) dx \\
 &= 4\pi \left[\ln(x-7) \right]_8^{10} \\
 &= 4\pi [\ln 3 - \ln 1] \\
 &= 4\pi \ln 3 \text{ unit}^3
 \end{aligned}$$

$$\begin{aligned}
 \text{e) } y &= \tan(x^2-3) \\
 \frac{dy}{dx} &= \sec^2(x^2-3) \cdot 2x \\
 &= 2x \sec^2(x^2-3)
 \end{aligned}$$

$$\begin{aligned}
 \int x \sec^2(x^2-3) dx &= \frac{1}{2} \int 2x \sec^2(x^2-3) dx \\
 &= \frac{1}{2} \tan(x^2-3) + C
 \end{aligned}$$

Question 3

$$\begin{aligned}
 \text{a) Prove} \\
 1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! \\
 = (n+1)! - 1
 \end{aligned}$$

$$\begin{aligned}
 \bullet \text{ for } n=1 \quad 1 \times 1! &= (1+1)! - 1 \\
 1 &= 2! - 1 \\
 &= 1
 \end{aligned}$$

true

$$\begin{aligned}
 \bullet \text{ assume true for } n=k \\
 S_k &= (k+1)! - 1
 \end{aligned}$$

$$\begin{aligned}
 \bullet \text{ prove true for } n=k+1 \\
 \text{i.e. Prove } S_{k+1} &= (k+2)! - 1
 \end{aligned}$$

$$\text{Now: } S_k + T_{k+1} = S_{k+1}$$

$$\begin{aligned}
 \therefore \text{LHS} &= (k+1)! - 1 + (k+1)(k+1)! \\
 &= (k+1)! [1 + (k+1)] - 1 \\
 &= (k+1)! (k+2) - 1 \\
 &= (k+2)! - 1 \\
 &= \text{RHS}
 \end{aligned}$$

• If true for $n=k$ then proved true for $n=k+1$.
Since true for $n=1$, must be true for $n=2$ and so on for all n .

$$\begin{aligned}
 \text{b) } \frac{\sin \theta}{\sin \theta + \cos \theta} - \frac{\sin \theta}{\sin \theta - \cos \theta} \\
 &= \frac{\sin \theta (\sin \theta - \cos \theta) - \sin \theta (\sin \theta + \cos \theta)}{(\sin \theta + \cos \theta)(\sin \theta - \cos \theta)} \\
 &= \frac{\sin^2 \theta - \sin \theta \cos \theta - \sin^2 \theta - \sin \theta \cos \theta}{\sin^2 \theta - \cos^2 \theta} \\
 &= \frac{-2 \sin \theta \cos \theta}{-\cos 2\theta} \\
 &= \frac{\sin 2\theta}{\cos 2\theta} \\
 &= \tan 2\theta
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } \sin^2 \theta &= \frac{1}{2} \sin 2\theta \\
 2 \sin^2 \theta &= \sin 2\theta \\
 2 \sin^2 \theta &= 2 \sin \theta \cos \theta \\
 \sin^2 \theta - \sin \theta \cos \theta &= 0 \\
 \sin \theta (\sin \theta - \cos \theta) &= 0 \\
 \sin \theta &= 0 \quad ; \quad \sin \theta = \cos \theta \\
 \tan \theta &= 1
 \end{aligned}$$

$$\therefore \theta = 0, \pi, 2\pi, \frac{\pi}{4}, \frac{5\pi}{4}$$

$$d) \quad p = \frac{2}{3} \quad q = \frac{1}{3}$$

$P(\text{at most 2 with brown eyes})$

$$P(X=2) + P(X=1) + P(X=0)$$

$$= {}^4C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^2 + {}^4C_1 \left(\frac{2}{3}\right)^1 \left(\frac{1}{3}\right)^3$$

$$+ {}^4C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^4$$

$$= 0.2963 + 0.0988 + 0.0123$$

$$= 0.41 \quad \left(\frac{33}{81}\right)$$

$$e) \quad y = \frac{x^2+1}{x}$$

$$y = x + \frac{1}{x}$$

$$\frac{dy}{dx} = 1 - x^{-2} \quad \left(1 - \frac{1}{x^2}\right)$$

$$\frac{d^2y}{dx^2} = 2x^{-3} \quad \left(\frac{2}{x^3}\right)$$

$$\text{Set points } \frac{dy}{dx} = 0$$

$$\therefore 1 - \frac{1}{x^2} = 0$$

$$x^2 - 1 = 0$$

$$x = \pm 1$$

$$x = 1$$

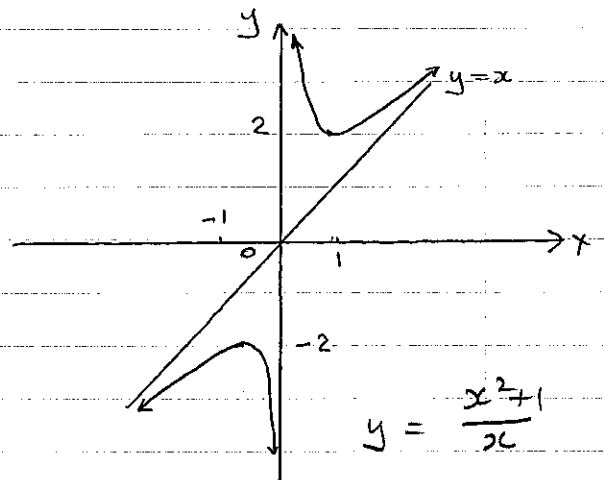
$$x = -1$$

$$y = 2$$

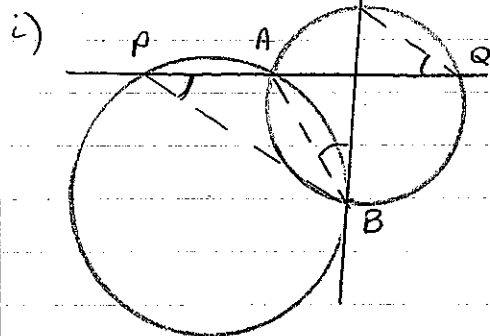
$$y = -2$$

Limits: as $x \rightarrow \infty$ $y \rightarrow x$
asymptotes $y = x$ and $x = 0$

No inflexion points as $\frac{2}{x^3} \neq 0$



Question 4



ii) $\angle RBA = \angle APB$ (angle in alternate segment)

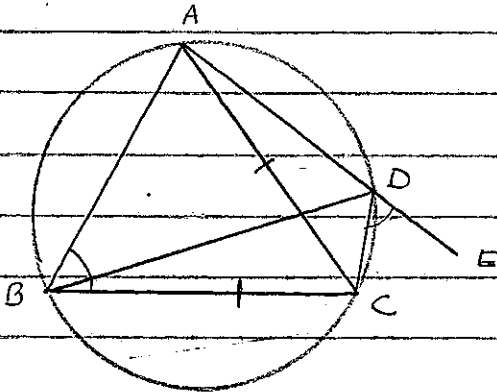
$\angle RBA = \angle AQR$ (angles standing on arc AR are equal)

$$\therefore \angle APB = \angle AQR$$

$\therefore PB \parallel QR$ (equal alternate angles $\angle APB, \angle AQR$)

b)

i)



$$\text{ii) } \angle ABC = \angle CDE$$

(exterior $\angle$ of cyclic quad equal to interior opposite $\angle$)

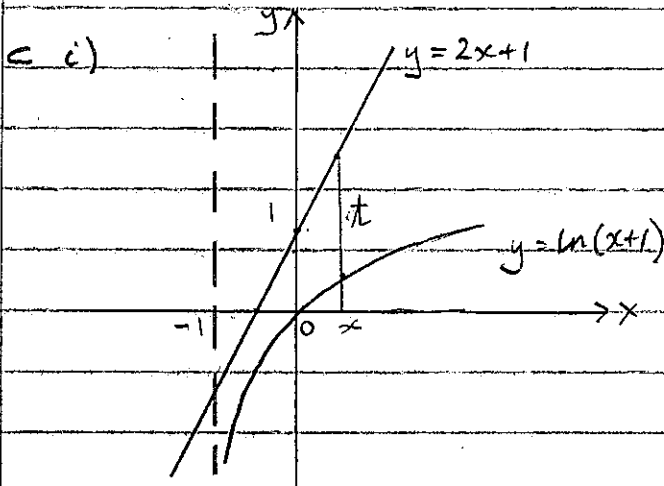
$$\text{iii) } \angle ABC = \angle BAC \text{ (equal } \angle\text{'s opposite equal sides of an isosceles triangle)}$$

$$\angle BAC = \angle BDC \text{ (angles standing on the same arc BC)}$$

$$\therefore \angle ABC = \angle BDC$$

$$\therefore \angle CDE = \angle BDC$$

$$\therefore DC \text{ bisects } \angle BDE$$



$$\text{ii) distance } t = 2x + 1 - \ln(x+1)$$

$$\frac{dt}{dx} = \frac{2 - 1}{x+1}$$

$$= 2 - (x+1)^{-1}$$

$$\frac{d^2t}{dx^2} = (x+1)^{-2}$$

$$= \frac{1}{(x+1)^2} \text{ (always +ve)}$$

$\therefore$ concave up

$$\text{When } \frac{dt}{dx} = 0$$

$$2 - \frac{1}{x+1} = 0$$

$$2(x+1) - 1 = 0$$

$$2x + 2 - 1 = 0$$

$$2x = -1$$

$$x = -\frac{1}{2}$$

$\therefore$ Distance between

curves is a min at $x = -\frac{1}{2}$